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SIT225: Data Capture Technologies

Activity 1.1: Arduino Blink

Welcome to Arduino!

Arduino is an electronic prototyping platform. Different types of sensors & actuators can be attached to Arduino boards to create our own sensing-thinking-acting systems.

Throughout this unit, we will use Arduino to create different sensing devices, and to retrieve the collected sensor data.

In this task, we will try out an introductory exercise, to learn the basic concepts of Arduino.

Hardware Required

- Arduino Board with in-built LED

- USB cable

Software Required

- Arduino programming environment

Pre-requisites: You must do the following before this task

Unit site weekly materials

Active learning sessions require reading material and/or videos available in the unit site, which you must read/view **BEFORE** you start the lab. If you are on-campus, this means that we expect you to have gone through these materials when you join active learning sessions.

Why should you read/watch pre-lab materials?

These materials will help you understand the background which the lab tasks require. Students come to university from diverse backgrounds. Some of you may be familiar with the background information, some of you may not. When you come to the lab prepared, you're already equipped

with confidence and will be able to participate in activities better. Ultimately, class time will be much more productive, dynamic, and fun for everyone.

Here are the pre-lab materials for our first task:

1. Watch TED Talk: <https://www.youtube.com/watch?v=UoBUXOOdLXY> (~15 minutes)
2. Read Arduino tutorial: <https://www.dummies.com/article/technology/computers/hardware/arduino/how-to-complete-your-first-arduino-sketch-164747>)
3. Read this task sheet from beginning to end.

Task Objective

“We have an Arduino board with an in-built LED light. We need the LED light to be turned on and off continuously, every one second.”

Activity Submission Details

Answer the questions below in this word document and other activities in this activity sheet to create a PDF and submit to OnTrack as described in this week's OnTrack task. PDFs of this activity sheet and OnTrack task need to be merged for submission in the OnTrack portal.

Q1: The TED talk given under the Pre-Lab materials, shows how Arduino is being used for interesting projects to capture data from the environment, process it, and use it to carry out useful actions.

Fill the given table below to answer the following:

What are **three** projects that use captured data as given in the TED talk? What data do they capture? What sensors do you think they could use to capture this data?

Project name	Data captured	Sensors to capture the data
Auto cat feeder	Cat chip distance	Distance sensor or chip detect sensor
Enough	Voice, TV single	Sound sensor, single sensor
Magic clock	Sound, vibration	Sound sensor, vibration sensor

Q2: Consider the given Task Objective. Think about how this simple system can be decomposed to ‘Sense-Think-Act’?

- a) What is the ‘sensing’ requirement in this system, if any?

The sensor require is the LED light.

- b) What is the ‘thinking’ requirement in this system, if any?

The “think” is think is the LED light turn off or on?

- c) What is the ‘acting’ requirement in this system, if any?

The acting is turn on or off the LED Light.

Q3: Please refer to the provided ‘Arduino Blink Activity Sheet’ and follow the steps.

- a) In Arduino-speak, what is a “sketch”?

In Arduino, Sketch mean the code or the instruction I put in IDE.

- b) setup() and loop() are key Arduino constructs. These are required in every Arduino sketch.

- i) Which of the above two, runs once at the very beginning of your program and never again (unless you reset or upload new code)?

Setup()

- ii) Which of the above two, is used to continuously run code over and over again?

Loop()

- c) What does **pinMode()** do?

Hint: <http://arduino.cc/en/Reference/HomePage>

pinMode() do set different pin in the Arduino to different mode such as input, output. So it read or output data.

- d) What is a comment?

To make a comment do “//”, after comment it will be ignore by the computer.

- e) What does the following line of code do:

`delay(x);`

Hint: <http://arduino.cc/en/Reference/HomePage>

It will stop program running for X sec.

- f) There is something you need to check before uploading your sketch. What is this?

Code error and COM port.

Q4: How can you test the Blink program to make sure it is working as given in the Task Objective?

First click the verify to check any error, if no error check the COM port is it connected correctly. If yes then click upload. Now we can see the LED is working.

Activity 1.2: Write Arduino data to serial communication port

Now you can blink Arduino's built-in LED, it is time to talk to outside Arduino-world, your computer which connects the Arduino board using a USB cable. Arduino IDE shows what you write to the serial port.

Hardware Required

Arduino Board with in-built LED

USB cable

Software Required

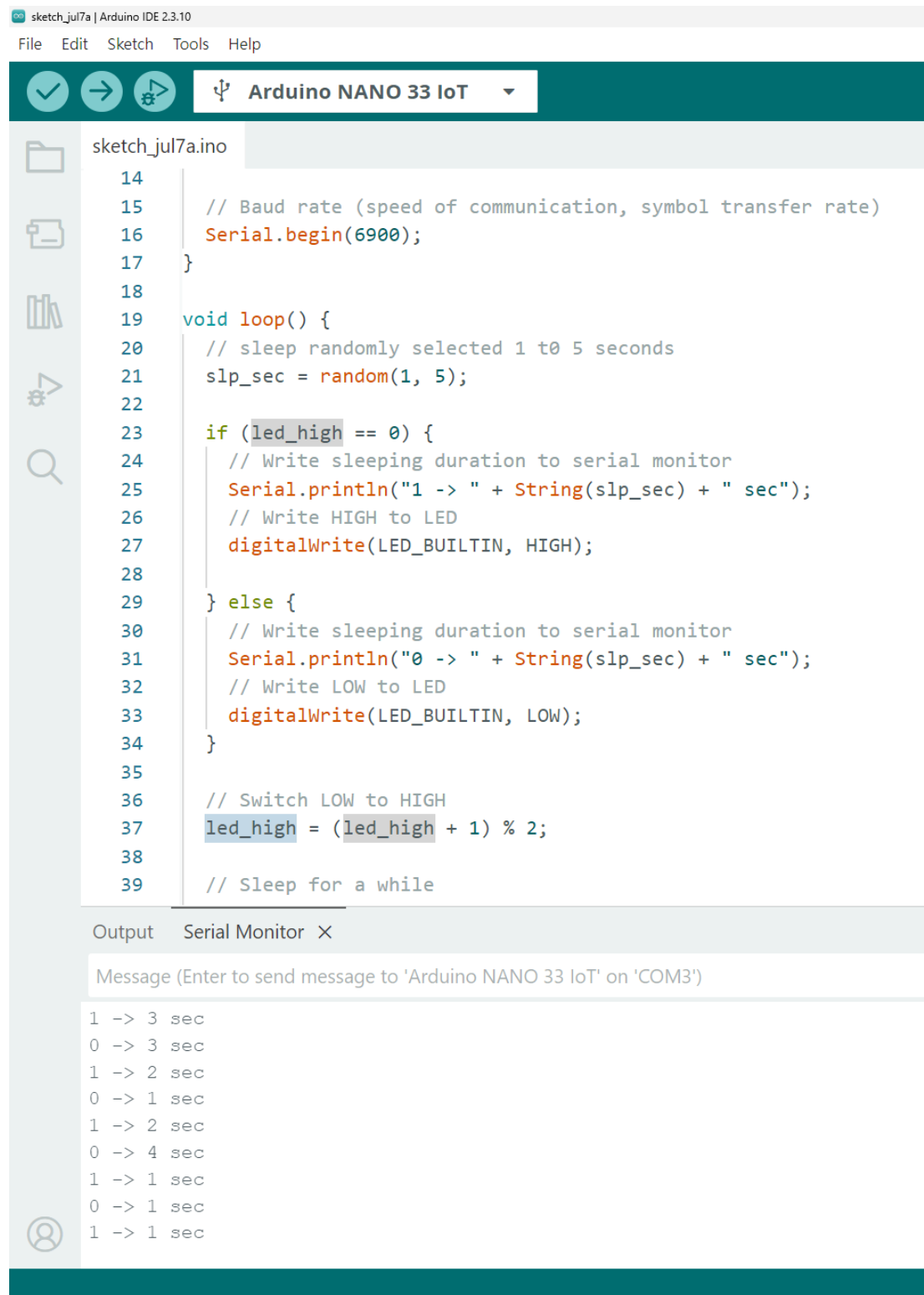
Arduino programming environment

Steps:

Steps	Actions
1	Identify the port through which Arduino is connected to your computer. You can find it in Arduino IDE Tools menu. Write Arduino sketch (or download from https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/sketch_blink.ino), which look like below, in Arduino IDE, deploy in your board and observe the output in IDE's serial monitor.

```
1  int led_high = 0; // variable to flag ON (1) or OFF (0)
2  int slp_sec = 0; // variable holds number of seconds to sleep
3
4
5  void setup() {
6      pinMode(LED_BUILTIN, OUTPUT); // set LED pin as output
7      digitalWrite(LED_BUILTIN, LOW); // switch off LED pin
8
9      // Baud rate (speed of communication, symbol transfer rate)
10     Serial.begin(6900);
11 }
12
13 void loop() {
14     // sleep randomly selected 1 to 5 seconds
15     slp_sec = random(1, 5);
16
17     if (led_high == 0) {
18         // Write sleeping duration to serial monitor
19         Serial.println("1 -> " + String(slp_sec) + " sec");
20         // Write HIGH to LED
21         digitalWrite(LED_BUILTIN, HIGH);
22     } else {
23         // Write sleeping duration to serial monitor
24         Serial.println("0 -> " + String(slp_sec) + " sec");
25         // Write LOW to LED
26         digitalWrite(LED_BUILTIN, LOW);
27     }
28
29     // Switch LOW to HIGH
30     led_high = (led_high + 1) % 2;
31
32     // Sleep for a while
33     delay(1000*slp_sec);
34 }
35
36
```

Question: Screenshot the serial monitor output and paste the image here.

Answer:

The screenshot displays the Arduino IDE 2.3.10 interface. The top menu bar includes File, Edit, Sketch, Tools, and Help. The toolbar shows icons for checking, running, and uploading code, along with a dropdown menu for the board, currently set to 'Arduino NANO 33 IoT'. The left sidebar contains icons for file explorer, serial monitor, and search. The main editor window shows the sketch 'sketch_jul7a.ino' with the following code:

```
14
15 // Baud rate (speed of communication, symbol transfer rate)
16 Serial.begin(6900);
17 }
18
19 void loop() {
20 // sleep randomly selected 1 to 5 seconds
21 slp_sec = random(1, 5);
22
23 if (led_high == 0) {
24 // Write sleeping duration to serial monitor
25 Serial.println("1 -> " + String(slp_sec) + " sec");
26 // Write HIGH to LED
27 digitalWrite(LED_BUILTIN, HIGH);
28
29 } else {
30 // Write sleeping duration to serial monitor
31 Serial.println("0 -> " + String(slp_sec) + " sec");
32 // Write LOW to LED
33 digitalWrite(LED_BUILTIN, LOW);
34 }
35
36 // Switch LOW to HIGH
37 led_high = (led_high + 1) % 2;
38
39 // Sleep for a while
```

Below the code editor is the 'Serial Monitor' window, which is titled 'Output Serial Monitor X'. It contains a text input field with the placeholder 'Message (Enter to send message to 'Arduino NANO 33 IoT' on 'COM3')'. The serial monitor displays the following output:

```
1 -> 3 sec
0 -> 3 sec
1 -> 2 sec
0 -> 1 sec
1 -> 2 sec
0 -> 4 sec
1 -> 1 sec
0 -> 1 sec
1 -> 1 sec
```

2	Question: Observe the use of “Serial” such as functions Serial.begin() in setup and Serial.println() in loop. Describe what these functions are doing with respect to the serial monitor output you have attached above. Answer: the above monitor output is how long it took to turn on and off the LED light. The Serial begin is to open up a communication channel between Arduino board and computer. The number mean Baud Rate, which is the speed of the communication. Serial.println sends text from the Arduino back to computer screen so I have output to see.
3	Question: If Arduino transfers data at 4800 bits per second (baud rate) and you're sending 12 bytes of data, how long does it take to send over this information? Answer: 1 byte = 8 bits 12 byte * 8 = 96 bits 96 bits / 4800bits = 0.02s

Activity 1.3: Arduino talks to Python

To listen to what Arduino sends, there will be a Python program running and keep listening to the same port where Arduino is writing data to receive it.

Hardware Required

Arduino Board with in-built LED

USB cable

Software Required

Arduino programming environment

Python 3.0 (Follow Python installation manual in unit site)

Steps:

Steps	Actions
1	Write Arduino sketch (or download from https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/sketch_serial_comm.ino) which looks like below. Open in Arduino IDE. Study the code. Upload the code to Arduino board and observe output in Arduino IDE serial monitor.

	<pre> 1 2 int x; 3 4 void setup() { 5 Serial.begin(9600); // set baud rate 6 } 7 8 void loop() { 9 while (!Serial.available()) {} // wait for data to arrive 10 11 // read string data from Serial, we know Python 12 // script is sending just an integer. 13 x = Serial.readString().toInt(); 14 15 // write a string (no newline) 16 Serial.print("Arduino sends: "); 17 18 // Add 1 to what received. 19 // Write an integer with a newline (println vs print). 20 Serial.println(x + 1); 21 22 // Push the data through serial channel. 23 Serial.flush(); 24 } </pre>
	Question: Do you see any output in serial monitor? If not, then why? Answer: this code is waiting user to input a number then Arduino turn input to int and do input + 1 then output. So it will appear any output at first. It will output only if I input a number.
2	Write Python code as below (or download from https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/serial_comm_script.py) and save it to a file serial_comm_script.py.

```
1  import serial
2  import random
3
4  # set baud rate, same speed as set in your Arduino sketch.
5  boud_rate = 9600
6
7  # set serial port as suits your operating system
8  s = serial.Serial('/dev/ttyACM1', boud_rate, timeout=5)
9
10 while True: # infinite loop, keep running
11
12     # a random number between 5 and 50.
13     data_send = random.randint(5, 50)
14
15     # write to serial port, set data encoding.
16     # Raw bytes are sent through serial ports, Python bytes() needs
17     # to know the encoding to generate bytes from string.
18     #
19     # We send a single integer which is read from Arduino sketch.
20     #
21     d = s.write(bytes(str(data_send), 'utf-8'))
22     print(f"Send >>> {data_send} ({d} bytes)")
23
24     # Read from serial port.
25     #
26     # readline keeps reading until a newline found in the data stream.
27     # Unlike write above, we just send an integer with no newline.
28     # You should receive data the same way as it is sent.
29     #
30     d = s.readline().decode("utf-8")
31     print(f"Recv <<< {d}")
32
```

```
serial_comm_script.py
C:\> Users > gordo > OneDrive > 桌面 > Deakin > Y3 > T2 > SIT225 Data Capture > Week 1 > serial_comm_script.py > ...
14 # s = serial.Serial('/dev/ttyACM1', baud_rate, timeout=5)
15 s = serial.Serial('COM3', baud_rate, timeout=5)
16
17 while True: # infinite loop, keep running
18
19     # a random number between 5 and 50.
20     data_send = random.randint(5, 50)
21
22     # write to serial port, set data encoding.
23     # Raw bytes are sent through serial ports, Python bytes() needs
24     # to know the encoding to generate bytes from string.
25     #
26     # We send a single integer which is read from Arduino sketch.
27     #
28     d = s.write(bytes(str(data_send), 'utf-8'))
29     print(f"Send >>> {data_send} ({d} bytes)")
30
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SERIAL MONITOR

gordo@MSI MINGW64 /c/Users/gordo/OneDrive/桌面/Deakin/Y3/T2/SIT225 Data Capture/Week 1
\$ py serial_comm_script.py

KeyboardInterrupt

gordo@MSI MINGW64 /c/Users/gordo/OneDrive/桌面/Deakin/Y3/T2/SIT225 Data Capture/Week 1
\$ py serial_comm_script.py

Traceback (most recent call last):
File "C:\Users\gordo\OneDrive\桌面\Deakin\Y3\T2\SIT225 Data Capture\Week 1\serial_comm_script.py", line 15, in <module>
s = serial.Serial('COM3', baud_rate, timeout=5)
File "C:\Users\gordo\AppData\Local\Programs\Python\Python312\Lib\site-packages\serial\serialwin32.py", line 33, in __init__
super(Serial, self).__init__(*args, **kwargs)
File "C:\Users\gordo\AppData\Local\Programs\Python\Python312\Lib\site-packages\serial\serialutil.py", line 244, in __init__
self.open()
File "C:\Users\gordo\AppData\Local\Programs\Python\Python312\Lib\site-packages\serial\serialwin32.py", line 64, in open
raise SerialException("could not open port {!r}: {!r}.format(self.portstr, ctypes.WinError())
serial.serialutil.SerialException: could not open port 'COM3': PermissionError(13, '存取被拒。', None, 5)

gordo@MSI MINGW64 /c/Users/gordo/OneDrive/桌面/Deakin/Y3/T2/SIT225 Data Capture/Week 1
\$

Run the Python file from command line using command:

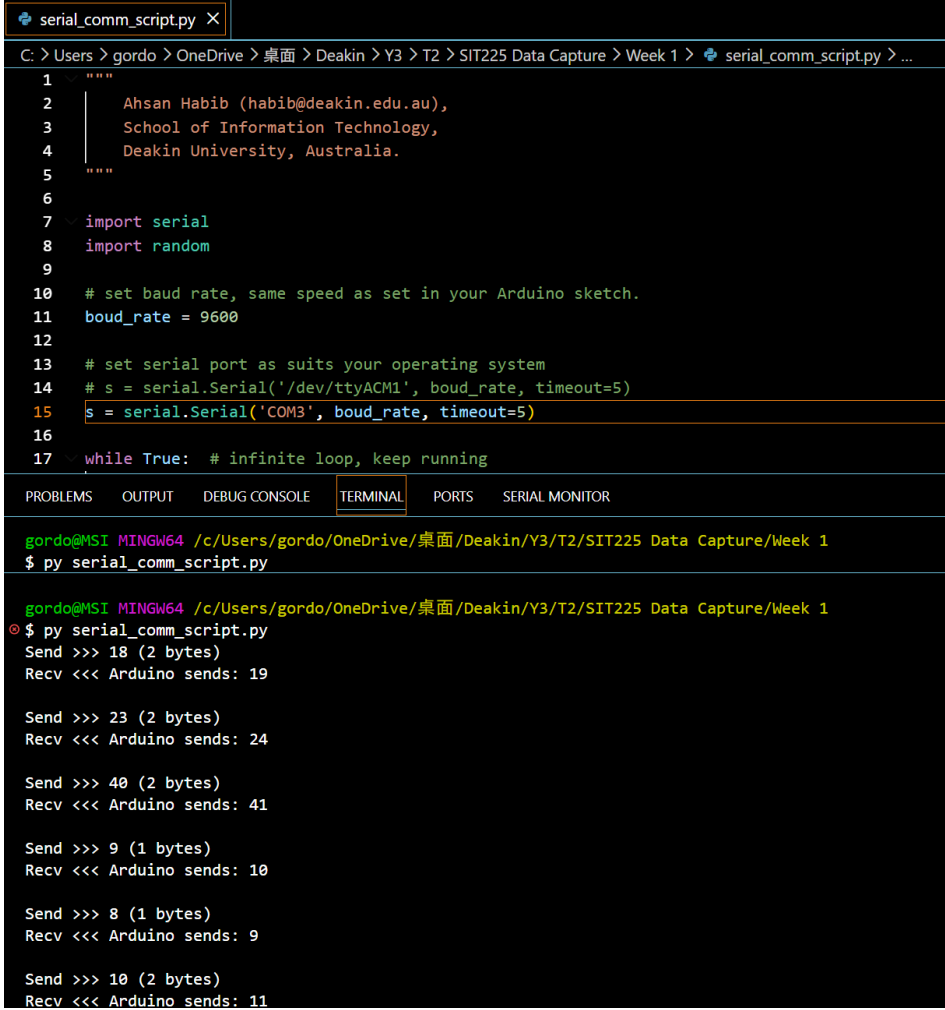
```
$ python serial_comm_script.py
```

Question: Run Python script in command line, does the script run or do you receive any error? If there is an error, analyse the error message and identify what went wrong.

Answer: "ModuleNotFoundError: No module named 'serial'" I receive this error, this appear because my python don't have a way to communicate with the Arduino not like Arduino IDE it have a built in function that can connect with the board. If I want to use python addition lib need to install.

After install the py -m pip install pyserial I get error 'could not open port '/dev/ttyACM1' this is because it not match my OS system so I change to COM3.

When I opening IDE, the py file will appear error “could not open port 'COM3'. This is because IDE is using or talking the board.

3	Question: Following the above step 2, do you think Arduino IDE is keeping the serial communication port busy talking to Arduino board? Now try to close the Arduino IDE and repeat step 2 above to run the Python script. Do previous errors show up again? If not, Python script should have print messages in the command line. Take the screenshot of the Python script output and paste here. Answer: “could not open port 'COM3' that the error I get when I open last IDE and try to run py file at the same time. So yes Arduino IDE is keeping serial communication port busy talking to Arduino board. After I close IDE, py file work.  <pre> 1 """ 2 Ahsan Habib (habib@deakin.edu.au), 3 School of Information Technology, 4 Deakin University, Australia. 5 """ 6 7 import serial 8 import random 9 10 # set baud rate, same speed as set in your Arduino sketch. 11 boud_rate = 9600 12 13 # set serial port as suits your operating system 14 # s = serial.Serial('/dev/ttyACM1', boud_rate, timeout=5) 15 s = serial.Serial('COM3', boud_rate, timeout=5) 16 17 while True: # infinite loop, keep running </pre> <pre> gordo@MSI MINGW64 /c/Users/gordo/OneDrive/桌面/Deakin/Y3/T2/SIT225 Data Capture/Week 1 \$ py serial_comm_script.py gordo@MSI MINGW64 /c/Users/gordo/OneDrive/桌面/Deakin/Y3/T2/SIT225 Data Capture/Week 1 \$ py serial_comm_script.py Send >>> 18 (2 bytes) Recv <<< Arduino sends: 19 Send >>> 23 (2 bytes) Recv <<< Arduino sends: 24 Send >>> 40 (2 bytes) Recv <<< Arduino sends: 41 Send >>> 9 (1 bytes) Recv <<< Arduino sends: 10 Send >>> 8 (1 bytes) Recv <<< Arduino sends: 9 Send >>> 10 (2 bytes) Recv <<< Arduino sends: 11 </pre>
4	Question: Observe the Python script output and describe the communication protocol used between Arduino sketch and Python script. Answer: looking at the code and output, this program first generate a random number and send to Arduino and let Arduino do x+1 math and send back the result and print out result. The port code in py code is a bit different because it use different system then IDE. /dev/ttyACM1 VS 'COM3'. They use same boud_rate

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